

4. Starch is made up of two molecules, amylose and amylopectin, shown in **Image 4.1**.

Image 4.1

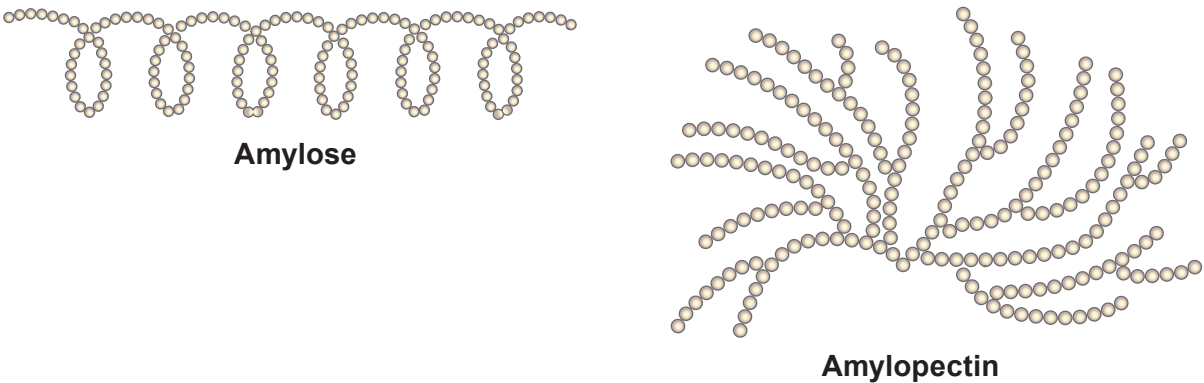
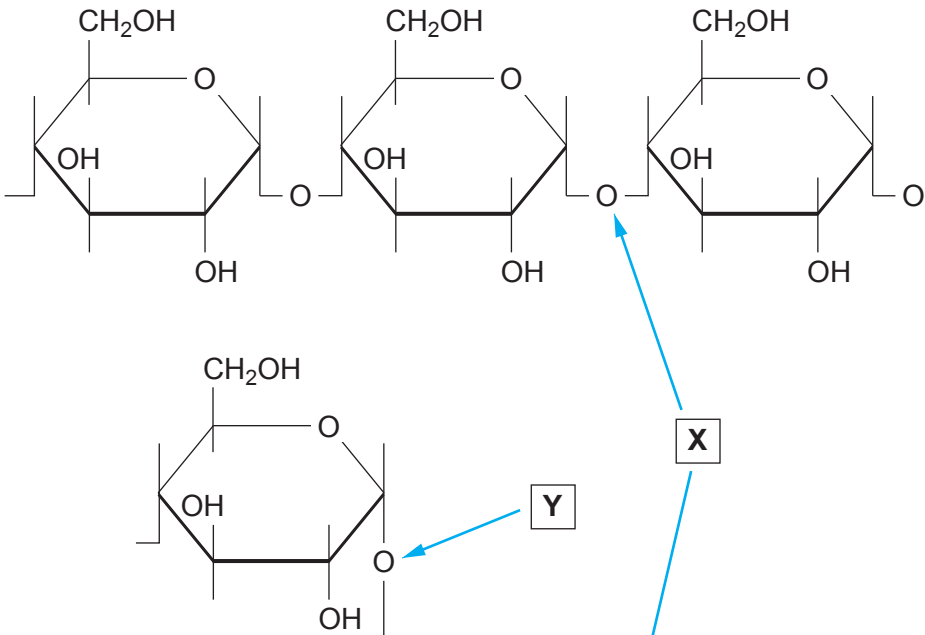


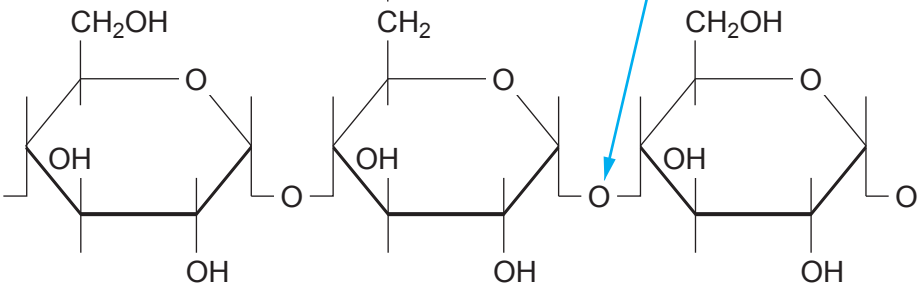
Image 4.2 shows sections of amylose and amylopectin.

Image 4.2

Amylose



Amylopectin



(a) (i) State the general name given to **bonds X** and **Y**.

[1]



(ii) State **one** similarity and **one** difference between amylose and amylopectin, shown in **Images 4.1** and **4.2**. [2]

Similarity

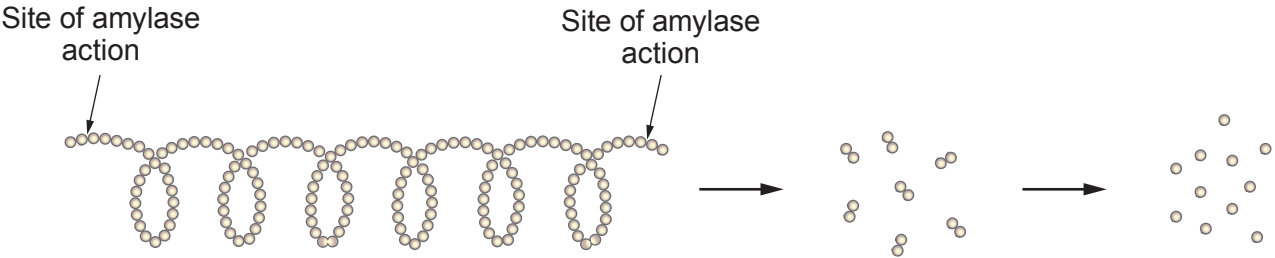
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Difference

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(b) In order to be absorbed into the blood, starch has to be hydrolysed to produce glucose. The digestion of amylose happens in two stages as shown in **Image 4.3**.

Image 4.3



(i) State what is meant by the term hydrolysis. [1]

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Amylopectin is digested in a similar way but needs an additional enzyme, isomaltase, for complete hydrolysis.

(ii) Explain the roles of the **three** enzymes involved in the complete hydrolysis of starch. [3]

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Examiner
only

(iii) One variety of corn used in animal feed has a lower amylose : amylopectin ratio than other varieties. With reference to **Images 4.2** and **4.3**, suggest why this lower ratio results in an increase in the rate of starch hydrolysis. [2]

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(iv) Describe a chemical test that could be used to show that starch is no longer present at the end of hydrolysis. [2]

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